

Description and choice data for the domain “Hotel rooms”

Description of the choice domain 28, Hotel rooms

The prompt question and the universe of five response options in the choice domain **Hotel rooms** are as follows. The labels *a*, *b*, *c*, *d* and *e* were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

Using Expedia results, I did a linear regression of price per night on a constant and the Expedia rating, in numbers of stars, for a sample of available hotels. The five levels of numbers of stars correspond roughly to the mean, plus and minus one sample standard deviation and plus and minus two standard deviations. Prices are approximately equal to fitted values in the linear regression.

Suppose you are staying over two nights in New York city. Which one of the following hotels would you choose, based on customer ratings and price per night?

- 3.6/5 stars, \$215 per night
- 3.9/5 stars, \$263 per night
- 4.2/5 stars, \$311 per night
- 4.5/5 stars, \$358 per night
- 4.8/5 stars, \$406 per night

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

Hotel rooms

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Figure 1: Screenshot for domain Hotel rooms

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	32	8	-	-	-	0.800	0.200	-	-	-
$\{a, c\}$	31	-	9	-	-	0.775	-	0.225	-	-
$\{b, c\}$	-	35	6	-	-	-	0.854	0.146	-	-
$\{a, b, c\}$	25	9	6	-	-	0.625	0.225	0.150	-	-
$\{a, d\}$	33	-	-	7	-	0.825	-	-	0.175	-
$\{b, d\}$	-	35	-	5	-	-	0.875	-	0.125	-
$\{a, b, d\}$	22	12	-	6	-	0.550	0.300	-	0.150	-
$\{c, d\}$	-	-	28	12	-	-	-	0.700	0.300	-
$\{a, c, d\}$	24	-	14	2	-	0.600	-	0.350	0.050	-
$\{b, c, d\}$	-	25	11	5	-	-	0.610	0.268	0.122	-
$\{a, b, c, d\}$	22	7	5	6	-	0.550	0.175	0.125	0.150	-
$\{a, e\}$	33	-	-	-	7	0.825	-	-	-	0.175
$\{b, e\}$	-	34	-	-	6	-	0.850	-	-	0.150
$\{a, b, e\}$	19	19	-	-	2	0.475	0.475	-	-	0.050
$\{c, e\}$	-	-	34	-	6	-	-	0.850	-	0.150
$\{a, c, e\}$	25	-	11	-	4	0.625	-	0.275	-	0.100
$\{b, c, e\}$	-	25	11	-	4	-	0.625	0.275	-	0.100
$\{a, b, c, e\}$	24	9	6	-	1	0.600	0.225	0.150	-	0.025
$\{d, e\}$	-	-	-	30	10	-	-	-	0.750	0.250
$\{a, d, e\}$	27	-	-	7	6	0.675	-	-	0.175	0.150
$\{b, d, e\}$	-	28	-	9	3	-	0.700	-	0.225	0.075
$\{a, b, d, e\}$	16	19	-	3	2	0.400	0.475	-	0.075	0.050
$\{c, d, e\}$	-	-	27	11	2	-	-	0.675	0.275	0.050
$\{a, c, d, e\}$	23	-	9	3	5	0.575	-	0.225	0.075	0.125
$\{b, c, d, e\}$	-	21	13	4	2	-	0.525	0.325	0.100	0.050
$\{a, b, c, d, e\}$	21	8	7	1	3	0.525	0.200	0.175	0.025	0.075

Table 1: Observed choice counts and proportions.

Choice data for domain 28, Hotel rooms

Table 1 shows choice counts and choice proportions for this choice domain. For each menu A and each object $x \in \{a, b, c, d, e\}$, $N_A(x)$ is the number of participants who chose object x from menu A and $\hat{P}_A(x)$ is the corresponding proportion of participants who chose x from A . When $x \notin A$, a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of $\{a, b, c, d, e\}$. The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

Warning in tritrafo(x, y, z): negative components

